

Alexander Maman

Molecular Genetics Department, Weizmann Institute of Science
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Education

- Ph.D. in Life Science, Weizmann Institute of Science** 2021 – Nov. 2026 (expected)
“Uncovering novel RNA modifications in extremophile Archaea.” Advisor: Prof. Schraga Schwartz
- M.Sc. in Biology, Weizmann Institute of Science** 2018 – 2021
“Exploring new functionalities and components of mRNA methylation.” Advisor: Prof. Schraga Schwartz
- B.Sc. in Agroecology & Plant Health, Hebrew University of Jerusalem** 2015 – 2018
Ein Prat Academy for Leadership & Maboia Philosophy Beit Midrash 2009 – 2010

Research & Academic Experience

- Graduate Researcher, Schwartz Lab, Weizmann Institute of Science** 2019 – present
Discovered the first natural RNA phosphate-backbone modifications (phosphorothioates). Developed specialized RNA-seq pipelines to detect RNA modifications. Identified new RNA modifying enzymes including the Esti writer family and its novel PAD domain through computational and functional approaches. Integrates wet-lab molecular genetics with computational biology.
- Yeast Genetics & Genomics Course, Cold Spring Harbor Laboratory, USA Summer 2022
- Research & Course Development Assistant, Weizmann Institute 2017 – 2018
- Lab Assistant, Prof. Moshe Coll's Insect Ecology Lab, Hebrew University 2016 – 2017

Publications

- **Maman**, Nobe, Fluke, ..., Schmitz, Santangelo, Taoka, Schwartz. “Phosphate Backbone Epitranscriptomics: Discovery of Natural RNA Phosphorothioates and their Writer Machinery.” *Cell* (in press).
- Ensink*, **Maman***, ..., Schwartz & van Werven. “The yeast RNA methylation complex consists of conserved yet reconfigured components with m6A-dependent and independent roles.” *eLife* 12 (2023): RP87860. (*Equal contribution)

Patents & Intellectual Property

“METHODS FOR CONTROLLING RNA POST-TRANSCRIPTION MODIFICATIONS.” US Provisional Application No. 64/000,421 (2026). Inventor together with Schraga Schwartz.

Research Funding

Proposals developed with Prof. Schraga Schwartz, based on my PhD work.

- **ISF MAPATZ- The Breakthrough Research Grant** (4 million ₪ (1.3 million \$), 5 years). “Backbone epitranscriptomics: Discovery, functional probing and mechanistic dissection of naturally occurring RNA phosphorothioates”
- **ERC Proof of Concept Grant** (€150,000). “STERNA – Enzymatic Stereospecific RNA Phosphorothioate Modification for Next-Generation Therapeutics.”
- **BINA Nurture Program** (\$232,000, 2 years). “Optimization of ESTI enzymes for precise PS modification of therapeutic RNAs.”

Awards

Azrieli Institute for Systems Biology Innovative Award

2022

Conference Presentations

Abstracts accepted; all 2026.

Short talk — EMBO: The Complex Life of RNA, Heidelberg Oct. 2026

Short talk — EMBO: Molecular mechanisms in evolution and ecology Oct. 2026

Short talk — tRNA 2026, Heraklion, Crete Sept. 2026

Short talk — RNA Modification and Processing 5th Symposium on Nucleic Acid Modifications, IMB Mainz Sept. 2026

Flash talk & poster — EMBL Conference: 10 Years of Tara Oceans, Heidelberg Sept. 2026

Academic Service

Ad-hoc reviewer (with Prof. Schraga Schwartz) for *Science*, *Nature Chemistry*, and *Nucleic Acids Research*.

Technical Skills

Computational: R, bash, basic Python. RNA-seq / NGS pipeline development. Phylogenetic and structural protein analysis (AlphaFold, PyMOL, in-house pipelines). Multi-agent AI workflows (Claude Code) for biological data exploration.

Molecular Biology: NGS protocol development. Advanced yeast genetics. Plasmid design & construction. RNA biochemistry. Basic tissue culture, lentivirus construction.

Research Vision

Pioneering “metaepitranscriptomics”: profiling RNA modifications across environmental microbial communities. Merging high-throughput sequencing, computational biology, and molecular experimentation to uncover evolutionary trends in RNA modification function across changing abiotic conditions, discover new RNA modifications, and document in-situ epitranscriptome dynamics as a potential biosensor.

Languages & Interests

Hebrew (native) | English (fluent) | Past “Hugay Sayarut” youth guide | Fantasy books enjoyer | Hydroponics initiate | John Muir admirer